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Project Based Learning

On

CIRCULAR WATER TANK DESIGN BY USING AUTOCAD.

By

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CIRCULAR WATER TANK DESIGN BY USING AUTOCAD

1. ABSTRACT

Acircular water tank is a critical reinforced concrete structure widely used for storing and distributing water across residential, commercial, agricultural, and industrial sectors [cite: 3]. This project focuses on the structural detailing and computer-aided drafting of a reinforced cement concrete (RCC) circular water tank utilizing AutoCAD [cite: 3]. Based on specific geometric constraints—including an inner diameter of 2040 mm, wall thickness of 230 mm, overall tank diameter of 2500 mm, total height of 3000 mm, and a 600 mm manhole—a comprehensive set of engineering drawings was produced [cite: 1, 2]. The drafting process encompasses plan views, sectional elevations (Section X-X'), foundation layout, Plain Cement Concrete (PCC) bedding details, and wall-footing structural connections [cite: 1, 2, 3]. AutoCAD tools were deployed to establish layer management, precise dimensioning, hatching, and standardized annotations to ensure clarity and professional presentation [cite: 3]. The resulting documentation provides a standardized reference model demonstrating how CAD technologies optimize accuracy and efficiency in structural engineering workflows [cite: 3].

2. INTRODUCTION

Water storage tanks arefundamental components of hydraulic infrastructure and municipal water supply systems [cite: 3]. Circular reinforced concrete tanks are structurally preferred due to their geometric efficiency [cite: 3]. The circular shape uniformly distributes hoop tension generated by internal hydrostatic water pressure, minimizing bending moments and allowing for optimal material usage compared to rectangular structures [cite: 3].

Computer-Aided Design and Drafting (CADD) using AutoCAD has replaced manual drafting in modern structural engineering practice [cite: 3]. AutoCAD allows engineers and CAD technicians to model structures with exact dimensional fidelity, implement standardized layering schemas, apply section hatching for material identification, and produce clear, reusable construction drawings [cite: 3]. This document presents a comprehensive CAD drafting methodology and structural report for a circular RCC water tank based on exact dimensional parameters extracted from plan and section engineering layouts [cite: 1, 2, 3].

3. OBJECTIVES OF THE PROJECT

- To prepare precise,scale-accurateCAD drawings of a circular reinforced concrete water tank based on given engineering dimensions [cite: 1, 2, 3].
- To draft detailed Plan Views and Sectional Elevation Views (Section X-X') illustrating structural components such as the RCC side walls, top slab, manhole opening, footing slab, and PCC base layer [cite: 1, 2].
- To implement professional CAD drafting standards, including structured layer management, precise line types, annotation styles, and dimensional formatting [cite: 3].
- To detail the foundation and sub-structure system, including the slab extension, depth below Ground Level (G.L), and plain cement concrete leveling layer [cite: 2, 3].
- To document essential AutoCAD commands, tools, and layer conventions used during the drafting process to serve as an academic and technical guideline [cite: 3].

4. PROJECT DESCRIPTION AND DESIGN DATA

The circularwater tank comprises an underground/partially elevatedreinforced concrete container structure with a circular plan layout set within a square slab/excavation boundary [cite: 1, 2]. The design parameters and precise geometric dimensions extracted from the structural drawings are summarized below [cite: 1, 2]:

Design Parameter / Structural Element	Dimension / Parameter Value	Description / Remarks
Internal Diameter of Tank ($Ø_{int}$)	2040 mm	Clear internal water storage diameter [cite: 1].
Outer Diameter of Tank ($Ø_{ext}$)	2500 mm	External boundary of RCC side walls [cite: 1].
Wall Thickness (t_w)	230 mm	Calculated as (Ø2500 - Ø2040)/2 [cite: 1].
Manhole Opening Diameter	600 mm (Ø600)	Inspection and maintenance access on top slab [cite: 1, 2].
Square Slab Base / Footing	4000 mm × 4000 mm	Main structural footing base

Design Parameter / Structural Element	Dimension / Parameter Value	Description / Remarks
Width		dimension [cite: 1, 2].
PCC Leveling Course Dimensions	4200 mm × 4200 mm	Base concrete course providing 100 mm offset [cite: 1, 2].
Total Tank Height (H_{total})	3000 mm	Height from top of footing slab to top roof slab [cite: 2].
Height Above Ground Level (G.L)	1000 mm	Freeboard and upper wall structure above ground [cite: 2].
Depth Below Ground Level (G.L)	2000 mm	Sub-surface tank body depth [cite: 2].
Top Slab Thickness	150 mm	Reinforced concrete cover slab [cite: 2].
Footing Slab Thickness	450 mm	Main base RCC structural raft/footing [cite: 2].
PCC Sub-base Thickness	100 mm	Plain Cement Concrete bed beneath footing [cite: 2].

5. DRAWING SET AND DETAILS

The CAD drawing set for the circular water tank comprises two primary representations [cite: 1, 2]:

A. Plan View

The Plan view depicts the top projection and internal concentric geometry of the tank structure [cite: 1]. It clearly details:

- The outer square boundary of the RCC footing (4000 mm × 4000 mm) [cite: 1].

- The outer square boundary of the underlying PCC bed (4200 mm × 4200 mm) [cite: 1].
- Concentric circular rings representing the outer wall surface (Ø2500 mm) and inner wall surface (Ø2040 mm) [cite: 1].
- An off-center 600 mm diameter circular manhole opening on the top slab [cite: 1, 2].
- Section line indicator **X - X'** passing through the center of the tank and cutting through the manhole [cite: 1].

B. Sectional Elevation (Section X-X')

The sectional elevation view illustrates the vertical profile and structural thickness of each component along cutting plane X-X' [cite: 1, 2]:

- **Sub-structure:** 100 mm thick PCC layer extending to a width of 4200 mm, supporting the 450 mm thick RCC footing slab of width 4000 mm [cite: 2].
- **Super-structure & Walls:** 230 mm thick RCC side walls rising 3000 mm high from the footing base [cite: 2].
- **Ground Elevation Interface:** Clear mark for Ground Level (G.L) located 2000 mm above the base footing line and 1000 mm below the top of the roof slab [cite: 2].
- **Roof Slab & Manhole:** 150 mm thick RCC roof slab featuring the 600 mm clear manhole opening at the top [cite: 2].

6. FOUNDATION DETAILS

The foundation system is engineered to safely transmit the self-weight of the RCC structure and hydrostatic water pressure into the underlying soil stratum [cite: 3].

- **Plain Cement Concrete (PCC) Sub-base:** A 100 mm thick PCC blinding layer (grade M10/M15) measuring 4200 mm × 4200 mm is provided to create a level, unyielding surface and prevent contamination of structural RCC during casting [cite: 2].
- **RCC Reinforced Base Footing:** A continuous rigid RCC slab footing (grade M30) of 450 mm thickness and 4000 mm × 4000 mm dimension rests directly over the PCC course [cite: 2]. The footing provides an edge projection beyond the 2500 mm diameter RCC tank walls, ensuring uniform load distribution [cite: 1, 2].
- **Wall-Footing Joint:** The 230 mm vertical RCC side walls are monolithically tied into the 450 mm footing slab using continuous starter bars and water-stops to form a watertight structural joint [cite: 2].

7. AUTOCAD DRAFTING METHODOLOGY

The step-by-step procedural methodology for drafting the circular water tank in AutoCAD includes [cite: 3]:

1. **Environment Setup:** Set drawing units to Millimeters (Decimal format) and configure drawing limits. Establish text styles, dimension styles, and layer schemes [cite: 3].
2. **Drafting the Plan View:**
   - Draw concentric circles using the CIRCLE command at a common center point for diameters Ø2040 mm and Ø2500 mm [cite: 1, 3].
   - Construct the square base boundaries (4000 mm and 4200 mm) using the RECTANG or LINE command aligned to the center [cite: 1, 3].
   - Position and draw the 600 mm manhole circle inside the inner wall ring [cite: 1, 3].
   - Draw the cutting section line X-X' passing through the diameter using line and arrow symbols [cite: 1].
3. **Drafting Section X-X':**
   - Project vertical construction reference lines down from the Plan View [cite: 3].
   - Draw horizontal baseline references for the 100 mm PCC bed and 450 mm base footing slab [cite: 2, 3].
   - Draw vertical RCC side walls extending 3000 mm above the footing base [cite: 2, 3].
   - Construct the 150 mm top roof slab including the 600 mm manhole opening break [cite: 2, 3].
   - Indicate Ground Level (G.L) at 2000 mm above the base footing line using standard earth hatch symbol patterns [cite: 2, 3].
4. **Hatching and Annotations:** Apply concrete hatch patterns (AR-CONC) for RCC elements and earth fill hatch for ground lines [cite: 3]. Add leaders, callouts, and explicit linear dimensioning [cite: 1, 2, 3].

8. AUTOCAD COMMANDS USED

The core AutoCAD tools and keyboard commands utilized in completing the structural drawing set include [cite: 3]:

Command	Shortcut	Application in Water Tank Project
CIRCLE	C	Drafting concentric tank walls (Ø2040, Ø2500) and manhole (Ø600) [cite: 1, 3].
LINE / PLINE	L / PL	Drawing footing profiles, section boundaries, and

Command	Shortcut	Application in Water Tank Project
		centerlines [cite: 1, 2, 3].
OFFSET	O	Creating parallel wall thickness lines, slab depths, and base margins [cite: 1, 2, 3].
TRIM / EXTEND	TR / EX	Cleaning intersections at wall joints, slab openings, and section cuts [cite: 3].
HATCH	H	Applying material patterns for RCC concrete walls/footing and ground earth [cite: 2, 3].
DIMLINEAR / DIMRADIUS	DLI / DRA	Adding vertical, horizontal, and radial measurement annotations [cite: 1, 2, 3].
MTEXT / LEADER	MT / LE	Creating section titles (SECTION X-X'), labels (PCC, FOOTING, G.L), and notes [cite: 1, 2, 3].
MOVE / COPY / MIRROR	M / CO / MI	Manipulating geometry and mirroring symmetrical wall section profiles [cite: 3].

9. RECOMMENDED AUTOCAD LAYER ORGANIZATION

Structured layering ensures drawing clarity, ease of modification, and standardized plotting [cite: 3]. Below is the recommended layer schema:

Layer Name	Color	Linetype	Lineweight	Description / Elements Placed
S-WALL	Cyan	Continuous	0.40 mm	RCC side walls and main structural concrete lines [cite: 2, 3].
S-FOOTING	Magenta	Continuous	0.35 mm	450 mm footing slab and top slab outlines [cite: 2, 3].
S-PCC	8 (Dark Gray)	Continuous	0.25 mm	100 mm PCC sub-base course boundary [cite: 2, 3].
S-GROUND	Green	Continuous / Dashed	0.25 mm	Ground Level (G.L) line and earth symbol [cite: 2].
S-ANNO-DIM	Red	Continuous	0.18 mm	All linear and radial dimension lines [cite: 1, 2, 3].
S-ANNO-TEXT	White / Yellow	Continuous	0.25 mm	Text annotations, callouts, titles,

Layer Name	Color	Linetype	Lineweight	Description / Elements Placed
				and labels [cite: 1, 2, 3].
S-HATCH	9 (Light Gray)	Continuous	0.13 mm	Concrete cross-section and soil hatch fill [cite: 2, 3].
S-CENTER	Red	CENTER2	0.18 mm	Centerlines and section cutting line X-X' [cite: 1, 3].

10. DESIGN AND DRAFTING CONSIDERATIONS

- **Hydrostatic Pressure Distribution:** Pressure increases linearly with depth ($p = \rho gh$). Walls must be drafted with sufficient thickness (230 mm) to resist maximum hoop tension at the base [cite: 2, 3].
- **Watertightness & Cracking:** Liquid retaining concrete requires high compaction, low water-cement ratio, and minimum crack-width design according to liquid storage codes [cite: 3].
- **Sub-surface Embedment:** Placing 2000 mm of the tank below Ground Level helps counteract thermal stresses and provides soil overburden resistance, while 1000 mm above ground facilitates easy gravity distribution or access [cite: 2].
- **Drafting Precision:** Object Snap (OSNAP) settings must be strictly active (Endpoint, Center, Intersection, Perpendicular) to avoid visual misalignment and dimensional inaccuracies in CAD [cite: 3].

11. DESIGN STANDARDS / REFERENCES

- **IS 3370 (Parts 1 - 4):** Indian Standard Code of Practice for Concrete Structures for the Storage of Liquids [cite: 3].
- **IS 456: 2000:** Code of Practice for Plain and Reinforced Concrete [cite: 3].

- **IS 875:** Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures [cite: 3].
- **AutoCAD User Documentation:** Autodesk CAD Standard Drafting Conventions for Civil and Structural Engineering [cite: 3].

12. APPLICATIONS

Circular reinforced concrete water tanks drafted using this CAD framework are applicable across multiple civil engineering domains [cite: 3]:

- Municipal water distribution networks and elevated/underground storage reservoirs [cite: 3].
- Residential housing complexes, apartments, and commercial facilities [cite: 3].
- Industrial process water storage, cooling tower basins, and effluent treatment plants [cite: 3].
- Agricultural irrigation reserves and rural water supply schemes [cite: 3].
- Dedicated emergency fire-water storage infrastructure [cite: 3].

13. ADVANTAGES OF AUTOCAD

- **High Precision and Accuracy:** Eliminates human scaling errors by drafting elements in 1:1 real-world millimeter dimensions [cite: 3].
- **Time Efficiency:** Commands like OFFSET, ARRAY, and MIRROR accelerate drawing generation compared to traditional manual drafting [cite: 3].
- **Layer Control:** Layer management allows seamless toggling of structural lines, annotations, and dimensions for specific construction views [cite: 3].
- **Ease of Revision:** Instant modification of wall thicknesses, depth specifications, or opening locations without redrawing the entire sheet [cite: 3].
- **Digital Interoperability:** Easy conversion to PDF format, digital archiving, and integration with structural analysis software [cite: 3].

15. RESULT

A complete 2D CAD structural drawing set of the RCC Circular Water Tank was successfully created using AutoCAD [cite: 3]. The generated plan and sectional elevation drawings accurately incorporate all given dimensions: inner diameter Ø2040 mm, outer diameter Ø2500 mm, total height 3000 mm, footing base 4000 mm × 4000 mm, PCC base 4200 mm × 4200 mm, and 600 mm manhole [cite: 1, 2]. Standard layer organization, hatch patterns, and dimension styles were applied to produce clear, professional structural construction sheets [cite: 3].

16. CONCLUSION

The project "Circular Water Tank Design by Using AutoCAD" demonstrates the effective

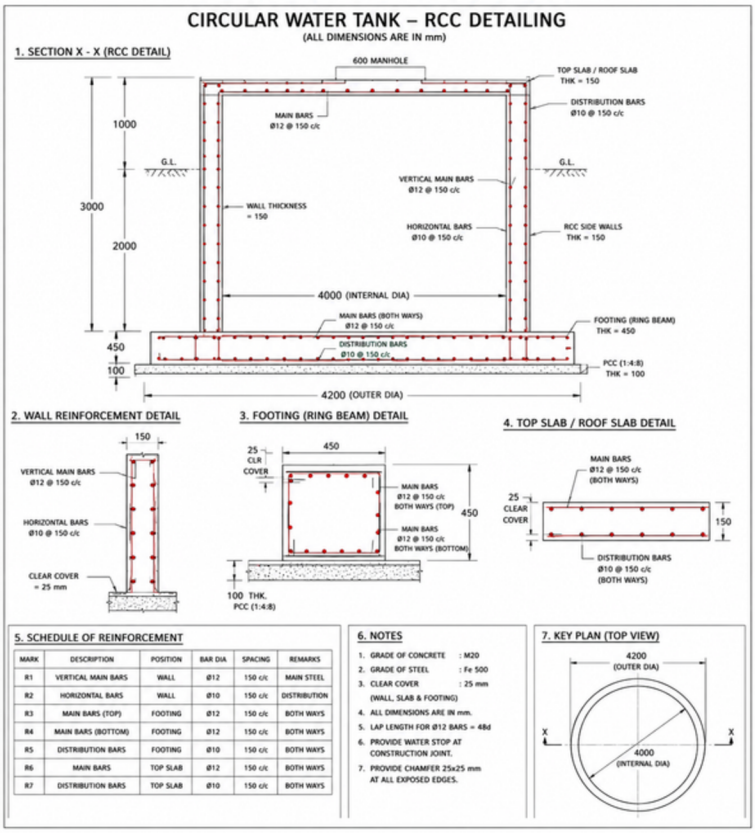
implementation of computer-aided drafting tools in civil and structural engineering [cite: 3]. By utilizing AutoCAD's drafting commands, layer organization, and dimensioning tools, structural details of the circular RCC tank—including its base footing, PCC sub-course, wall thickness, and top slab access—were represented with complete precision [cite: 1, 2, 3]. This documentation serves as a practical academic and professional reference model for structural detailing and computer-aided engineering graphics [cite: 3].

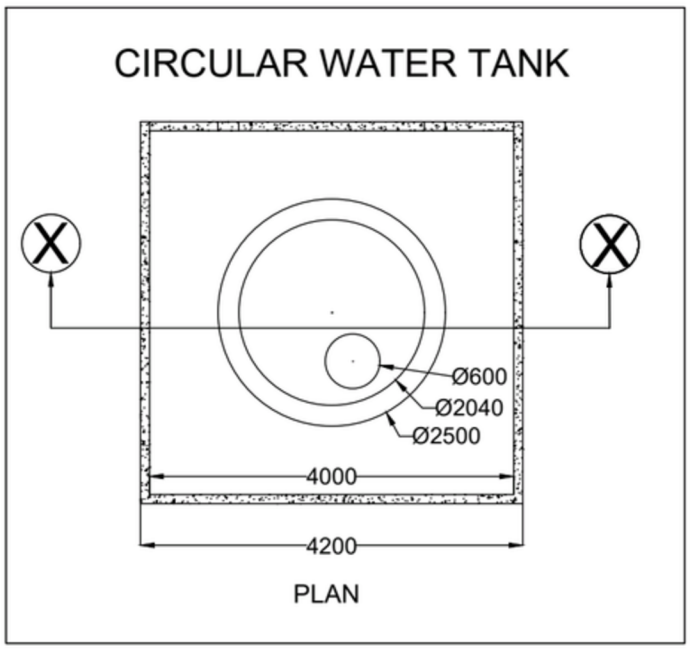
17. LEARNING OUTCOMES

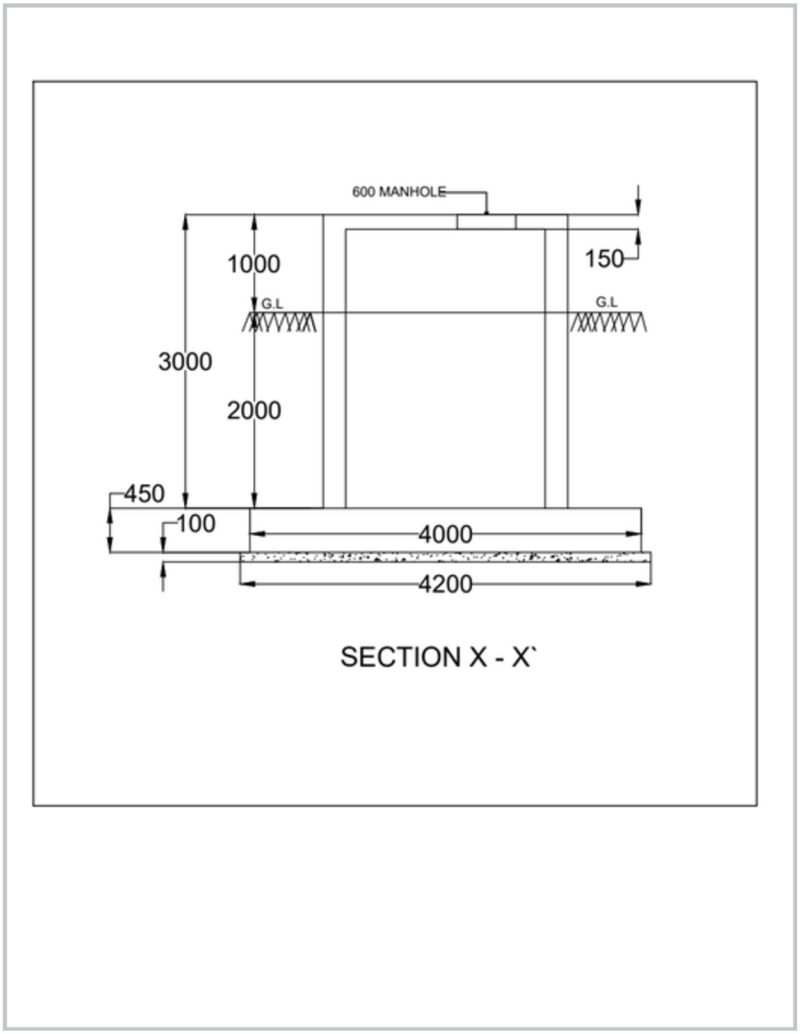
- Mastered the application of AutoCAD commands (CIRCLE, OFFSET, HATCH, TRIM) for structural drafting [cite: 3].
- Gained practical knowledge in translating physical engineering dimensions and sectional cuts into 2D technical drawings [cite: 1, 2, 3].
- Understood structural components of liquid-retaining structures, including PCC leveling beds, RCC raft footings, side walls, and roof slabs [cite: 2, 3].
- Developed competency in establishing CAD layer standards, dimension styles, text annotations, and hatching conventions [cite: 3].

18. REFERENCES

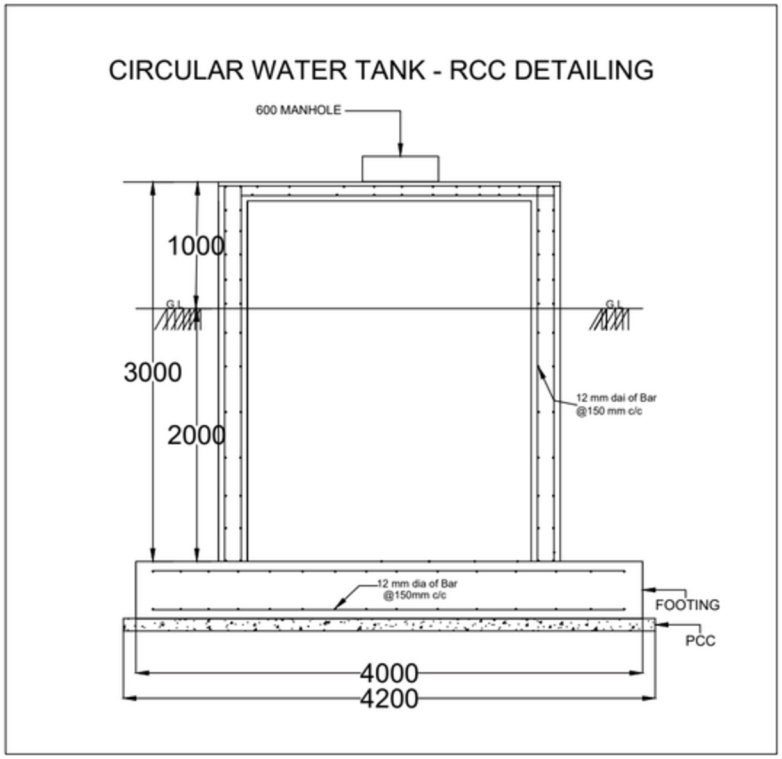
1. Structural Engineering Drawing Sheet: *CIRCULARWATER TANK Plan View (0600, 02040, 02500, 4000, 4200)* [cite: 1].
2. Structural Engineering Drawing Sheet: *CIRCULARWATER TANK Section X - X' (3000, 2000,1000,G.L,450Footing,100PCC,150Slab,600Manhole)* [cite: 2].
3. Academic Project Report: *CircularWaterTankDesignby Using AutoCAD*, Prepared for Engineering Drafting Project [cite: 3].
4. IS 3370 (Part I & II): *IndianStandardCodeofPracticefor Concrete Structures for the Storage of Liquids*, Bureau of Indian Standards [cite: 3].
5. IS 456: 2000: *Plain and Reinforced Concrete - Code of Practice*, Bureau of Indian Standards [cite: 3].


CIRCULAR WATER TANK – RCC DETAILING
(ALL DIMENSIONS ARE IN mm)
1. SECTION X - X (RCC DETAIL)
600 MANHOLE
TOP SLAB / ROOF SLAB THK = 150
DISTRIBUTION BARS Ø10 @ 150 c/c
MAIN BARS Ø12 @ 150 c/c
1000
3000
2000
G.L.
WALL THICKNESS = 150
VERTICAL MAIN BARS Ø12 @ 150 c/c
HORIZONTAL BARS Ø10 @ 150 c/c
RCC SIDE WALLS THK = 150
4000 (INTERNAL DIA)
MAIN BARS (BOTH WAYS) Ø12 @ 150 c/c
FOOTING (RING BEAM) THK = 450
DISTRIBUTION BARS Ø10 @ 150 c/c
450
100
PCC (1:4:8) THK = 100
4200 (OUTER DIA)
2. WALL REINFORCEMENT DETAIL
150
VERTICAL MAIN BARS Ø12 @ 150 c/c
HORIZONTAL BARS Ø10 @ 150 c/c
CLEAR COVER = 25 mm
3. FOOTING (RING BEAM) DETAIL
450
25 CLR COVER
MAIN BARS Ø12 @ 150 c/c BOTH WAYS (TOP)
MAIN BARS Ø12 @ 150 c/c BOTH WAYS (BOTTOM)
450
100 THK. PCC (1:4:8)
4. TOP SLAB / ROOF SLAB DETAIL
MAIN BARS Ø12 @ 150 c/c (BOTH WAYS)
25 CLEAR COVER
150
DISTRIBUTION BARS Ø10 @ 150 c/c (BOTH WAYS)
5. SCHEDULE OF REINFORCEMENT
MARK | DESCRIPTION | POSITION | BAR DIA | SPACING | REMARKS
R1 | VERTICAL MAIN BARS | WALL | Ø12 | 150 c/c | MAIN STEEL
R2 | HORIZONTAL BARS | WALL | Ø10 | 150 c/c | DISTRIBUTION
R3 | MAIN BARS (TOP) | FOOTING | Ø12 | 150 c/c | BOTH WAYS
R4 | MAIN BARS (BOTTOM) | FOOTING | Ø12 | 150 c/c | BOTH WAYS
R5 | DISTRIBUTION BARS | FOOTING | Ø10 | 150 c/c | BOTH WAYS
R6 | MAIN BARS | TOP SLAB | Ø12 | 150 c/c | BOTH WAYS
R7 | DISTRIBUTION BARS | TOP SLAB | Ø10 | 150 c/c | BOTH WAYS
6. NOTES
1. GRADE OF CONCRETE : M20
2. GRADE OF STEEL : Fe 500
3. CLEAR COVER : 25 mm (WALL, SLAB & FOOTING)
4. ALL DIMENSIONS ARE IN mm.
5. LAP LENGTH FOR Ø12 BARS = 48d
6. PROVIDE WATER STOP AT CONSTRUCTION JOINT.
7. PROVIDE CHAMFER 25x25 mm AT ALL EXPOSED EDGES.
7. KEY PLAN (TOP VIEW)
4200 (OUTER DIA)
X
X
4000 (INTERNAL DIA)


CIRCULAR WATER TANK
Ø600
Ø2040
Ø2500
4000
4200
PLAN



SECTION X - X`


CIRCULAR WATER TANK - RCC DETAILING
600 MANHOLE
1000
G.L
G.L
3000
2000
12 mm dai of Bar
@150 mm c/c
12 mm dia of Bar
@150mm c/c
FOOTING
PCC
4000
4200